

LITERATURE SURVEY

AI-Based Image-to-Recipe Generation System Using CNN and Sequence Modeling

3. Literature Survey

The proposed AI-Based Image-to-Recipe Generation System aims to generate cooking recipes directly from food images by combining Convolutional Neural Networks (CNNs) for ingredient recognition with sequence generation models for recipe creation. This literature survey reviews previous research, existing image recognition techniques, recipe generation models, enabling technologies, system limitations, research gaps, and explains how the proposed system improves existing approaches.

3.1 Introduction

Artificial Intelligence, Computer Vision, and Natural Language Processing have enabled automatic understanding of food images. Modern deep learning models can recognize dishes and predict ingredients with high accuracy. However, converting visual information into complete cooking recipes remains a challenging task because recipes require logical ingredient ordering, cooking methods, and contextual understanding.

3.2 Review of Existing Research

Paper 1: Bag-of-Features Food Recognition using handcrafted image descriptors.

Paper 2: CNN-based Food Classification using ResNet and EfficientNet.

Paper 3: Im2Recipe embedding models matching food images with recipes.

Paper 4: Inverse Cooking framework using CNN and LSTM.

Paper 5: Attention-based ingredient extraction.

Paper 6: Knowledge graph assisted recipe generation.

Paper 7: Vision Transformer with language models for coherent recipe generation.

Paper 8: Reinforcement learning for improving recipe quality.

3.3 Existing Commercial Solutions

Applications such as Google Lens, Yummly and Tasty allow users to search recipes or identify dishes. However, most depend on existing recipe databases instead of dynamically generating recipes from visual input.

3.4 Technologies Used

Python, PyTorch, OpenCV, Flask, HTML, CSS, JavaScript, pretrained CNN models, sequence generation models, and publicly available food datasets.

3.5 Limitations of Existing Systems

Most systems depend on fixed recipe databases, struggle with mixed dishes, provide limited personalization, and cannot accurately estimate ingredient quantities.

3.6 Research Gap

Few existing systems combine image recognition, ingredient prediction, and intelligent recipe generation into one integrated solution using modern deep learning.

3.7 Proposed System Improvements

The proposed system integrates image preprocessing, CNN-based ingredient recognition, sequence-based recipe generation, and an easy-to-use web interface. Future enhancements include nutritional analysis, multilingual recipes, and personalized recommendations.

3.8 Comparative Analysis

Compared with traditional recipe search systems, the proposed solution provides end-to-end recipe generation directly from uploaded food images.

3.9 Conclusion

The literature indicates that AI-powered food understanding has advanced significantly. The proposed Image-to-Recipe Generation System provides a practical, scalable, and cost-effective solution using modern deep learning and open-source technologies.